Bachelor’s Theses

(SS26 – Academic Year 2025/2026),

Spring Semester SS26

Orientation & Guidelines

February 2026

Ahmed M. H. Abdelfattah

Media Engineering and Technology

(German University in Cairo)

Outline

- **Terms & Conditions** (business rules)
- **Course**(?) **Format**
- **Assessment** & Grading
- **Follow-up** procedure
- Initial **Tasks**
- **Scientific Papers**
- SS26 **Topics & Clusters**

About: Terms & Conditions

- The engineering bachelor thesis is **an individual piece of undergraduate research that has a separate grade/scale/transcript.**

- It may involve:
  - the design and implementation of some complex computational system,
  - the construction and evaluation of a useful tool or a novel solution to an interesting problem, or
  - a thorough/insightful review of the literature on a particular topic.

- To successfully complete your thesis work you are expected to be diligent, hardworking, committed, and independent.

- To achieve the highest grade, you must adhere to the rules (إعمل الصح). ☺

Course(?) Format

- **Development and Management** (*iAgile*)
  - Work Plan & Gantt Chart, Trello, Google Drive, Overleaf, etc.
- Weekly **Meetings**
  - Meeting Minutes (must be taken by the student and then shared)
- Weekly **Deliverables** *(tangibles/products)*:
  - E.g., Short presentation, Video, Report?, Implementation, Results
- **Tools**
  - Python, Java, Matlab, Mathematica, LaTeX/Overleaf, etc.
  - Gen-AI Tools *(important: Check the shared GUC Policy)*
- **Thesis** and **Oral Examination** / Thesis rehearsal

B.Sc. Spring 2026 Timeline - Round I

OFFICIAL START
Saturday, February 14, 2026

TOTAL DURATION:
Approx. 114 DAYS
(16 WEEKS)

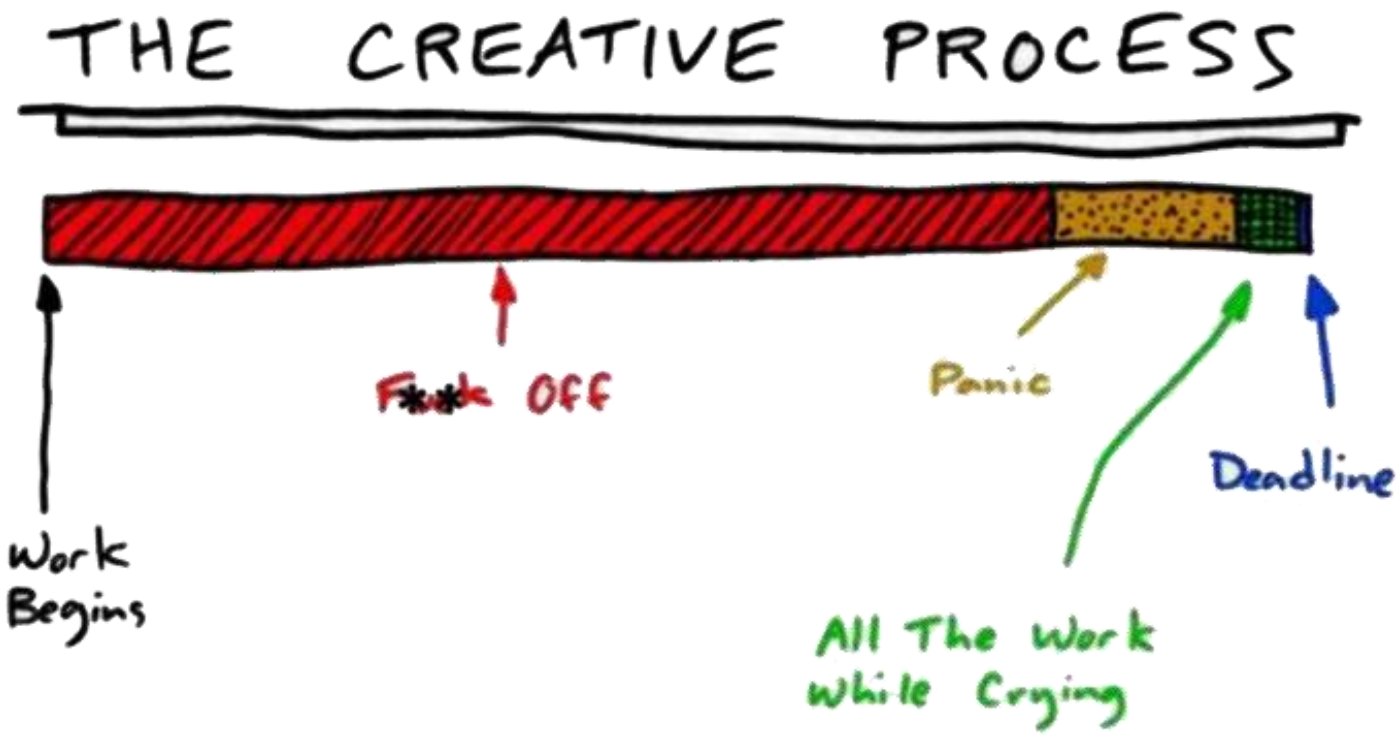
SUBMISSION DATE
Monday, June 8, 2026

Spring Final Exams: June 1-17 | Presentations: June 20 - July 2 | Final Version Deadline: Aug 31

B.Sc. Timeline [Spring2026] – Round I	
B.Sc. Official starting date	Saturday, February 14, 2026
B.Sc. Submission date	Monday, June 8, 2026
B.Sc. Presentation	Saturday, June 20 to, Thursday, July 2,2026
Spring Final Exams 2026	Monday, June 1st to Wednesday, June 17, 2026
Deadline to submit final B.Sc. version (after comments)	Monday, August 31, 2026

Timeline: *Are you Exceptional???*

We'll see!



Ever heard of "time management"?
It is the world's most precious non-renewable resource, yet often the most ignored.
Ultimately, we are like the (m)onkey: we only learn to swim once the water reaches our ears.
Don't wait for the flood to find your stroke.

Meetings:

- You should have a weekly meeting with your supervisor.
- You should not miss more than 4 meetings without an approved excuse. Otherwise, you are risking failing the bachelor thesis.

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- You should have a weekly meeting with your supervisor.
- You should not miss more than 4 meetings without an approved excuse. Otherwise, you are risking failing the bachelor thesis.
- You should come prepared for the meeting. This includes giving a full (oral) report on your progress over the week and a concrete description of challenges or problems encountered.

Thesis Report:

- You should write your thesis report using LaTeX.
  - Start learning LaTeX immediately, if you are not familiar with it.
  - You can use the Overleaf online LaTeX editor (`www.overleaf.com`).
- You should follow the MET bachelor thesis template that you receive from your supervisor. https://www.overleaf.com/read/fbkbcmqgmfnn
- Start writing your thesis report immediately. Every week you will be assigned some tasks; you should write about your progress in these tasks weekly. Hence, the thesis report is expected to be updated by the week.

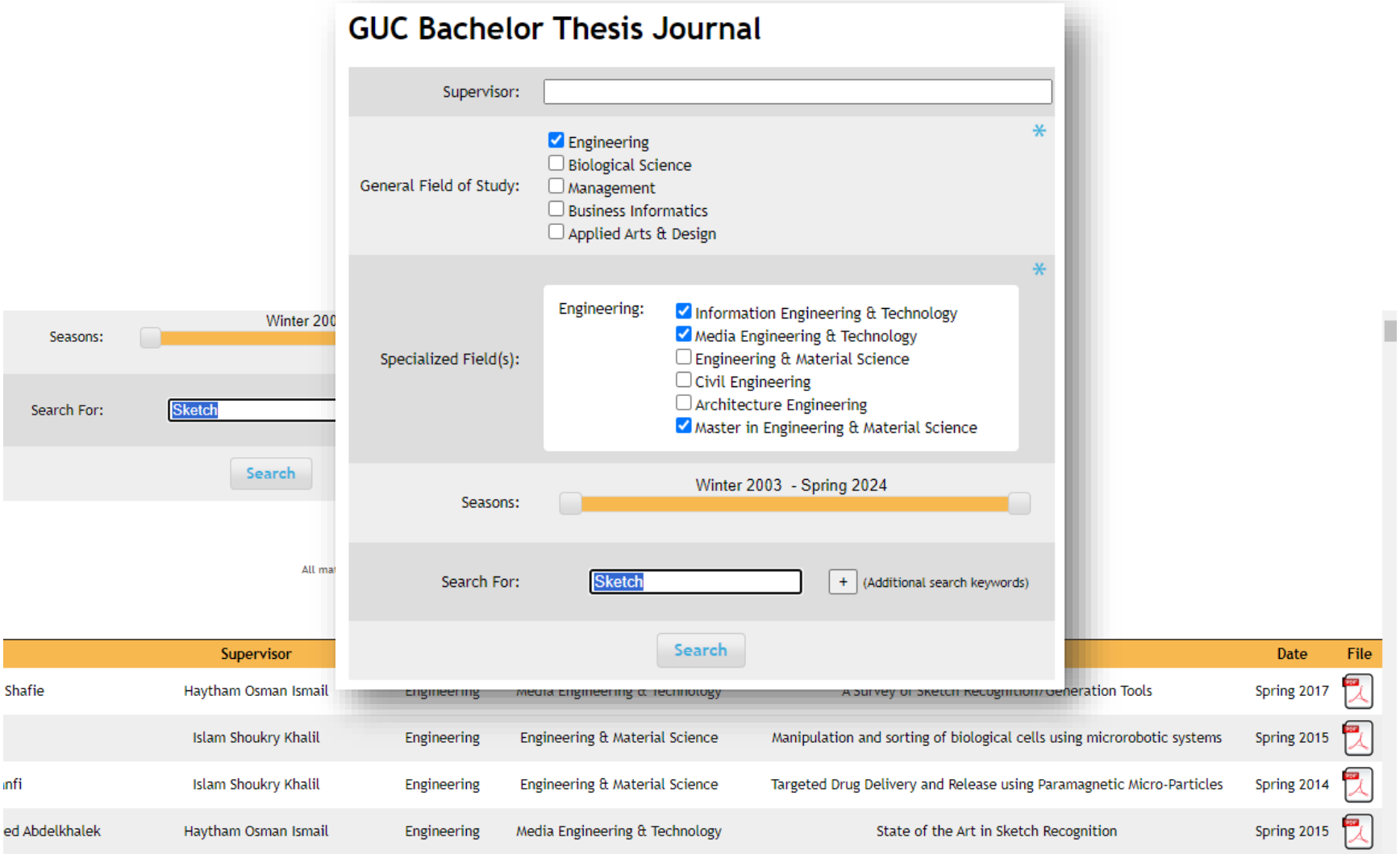
Thesis Report:

- Overleaf template available HERE. https://www.overleaf.com/read/fbkbcmqgmfnn
- Or, download the complete template of a GUC thesis from this link:
  - Link: https://drive.google.com/file/d/1M9IVYbNBef2zV-f_fz6wQ57JxNENxEfI/view?usp=drive_link
- Your reports eventually constitute part of your thesis. It's advisable, thus, to **start filling-in** your own thesis template with your own reports and everything else.

- *Introducing yourself to the LaTeX typesetting system takes one or two days*
  - *(do this introduction today! Now! It's easy!), [Why LaTeX?]*
  - *then do everything else tomorrow (or today after learning ☺)*

- *(next slide)* View GUC theses at: https://researchjournal.guc.edu.eg/search.aspx

- View GUC Bachelor theses at: https://researchjournal.guc.edu.eg/search.aspx



Assessment:

Progress and continuous evaluation	40%
(Supervisor) **Presentation**	15%
(Supervisor) **Thesis**	15%
(External Reviewer) Presentation	*15%*
(External Reviewer) Thesis	*15%*

70% Supervisor

30% External Reviewer

Grading:

– A+	94	– B+	80	– C+	60
– A	90	– B	75	– C	55
– A-	85	– B-	67	– C-	50

Assessment:

	Max Mark
Report presentation	
English, spelling, punctuation, and style	5
Logical and orderly layout of report	5
Development, achievement, and conclusion	
Analysis of all factors in the problem,	5
Oral presentation	
Presentation technical content and discussion	5
Use of language	5
Presentation Quality	5
Total Mark	30

	Max Mark
Quality of interim reports	10
Presentations and discussion	10
Quality of work	15
Independency	5
Total Mark	40

High grade: (85%–89%, A-) To give a grade in this range, the project must satisfy some specific characteristics.

1. The student fulfilled all the project aims.
2. The student clearly appreciate the project's applications.
3. The student conducted the work according to sound professional practices.
4. The project has sound theoretical or analytical basis.
5. Good written report.
6. Good oral presentation.

Grade	Mark Range Reference
A+	94 – 100
A	90 – 93
A-	85 – 89
B+	80 – 84
B	75 – 79
B-	67 – 74
C+	60 – 66
C	55 – 59
C-	50 – 54
F	49 and below

The gratitude of a donkey is a kick!

Assessment Guidelines:

- **Attention to detail**
  - Small details impact the overall quality.
  - Do not attack/outsmart business rules!

- **Constructive approach**
  - Approach the challenges by promoting a constructive and thoughtful attitude.
  - *Is self-learning suitable for everyone?*

- **Individual responsibility**
  - LLMs are NOT friends! (but useful)
  - Take personal responsibility for comprehending the content you encounter.

    … all content! *(again: Check the shared GUC Policy for using Gen-AI)*

- **Supervision vs. Teaching**
  - Follow active and engaging approaches to the learning process (this is not passive).
  - Know yourself https://www.16personalities.com/ *(ACHTUNG: You're dealing with a 100% INTJ-T)*

(main) Deliverables:

Deliverables other than the final thesis are usually due on the day of your weekly meeting as hard copies.

Milestone deliverables:

- **Semester Plan:**
  - A (tentative/progressive) week-by-week plan for weekly achievements
  - During GUC Week TW2 (Deadline: 20 February)

- **1st interim report:**
  - The thesis report as of the day of submission.
  - By 19-March-2026 @11:59pm CLT.

- **2nd interim report:**
  - Again, this is your thesis so-far.
  - By 07-May-2026 @11:59pm CLT.

- **Rehearsal presentations:**
  - Starting Revision Week 19-May-2026 (more will be shared later)
- **Finalized thesis:**
  - 08-June-2026

Softcopies or/and files shared online are NOT deliverables (yet are mandatory!).

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Next week???

Softcopies or/and files shared online are NOT deliverables (yet are mandatory!).

(general) Deliverables:

- Objectives should reflect all deliverables that may result from your project, not only the final application. These should match the problem statement and formulated as SMART objectives:

  Specific, Measurable, Achievable, Relevant, and Time-Bound.

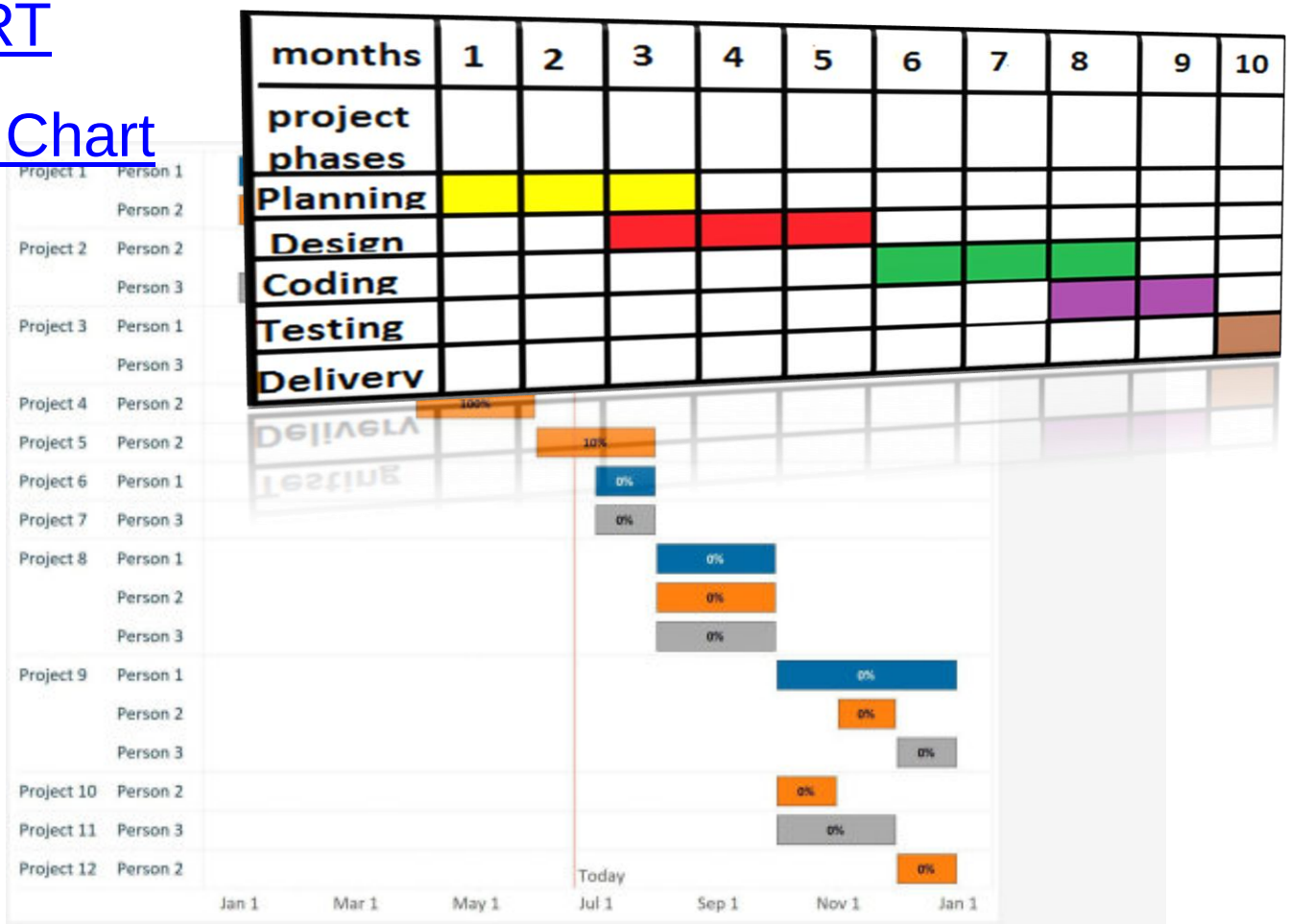
- *Readings:*
  - How to write SMART goals?
  - Writing Meaningful Goals and SMART Objectives

- SMART objectives
- A SMART objective is one that is *specific*, *measurable*, *achievable*, *relevant*, and *time-bound*.

https://www.health.state.mn.us/communities/practice/resources/phqitoolbox/objectives.html#:~:text=A%20SMART%20objective%20is%20one,and%20designing%20a%20work%20plan

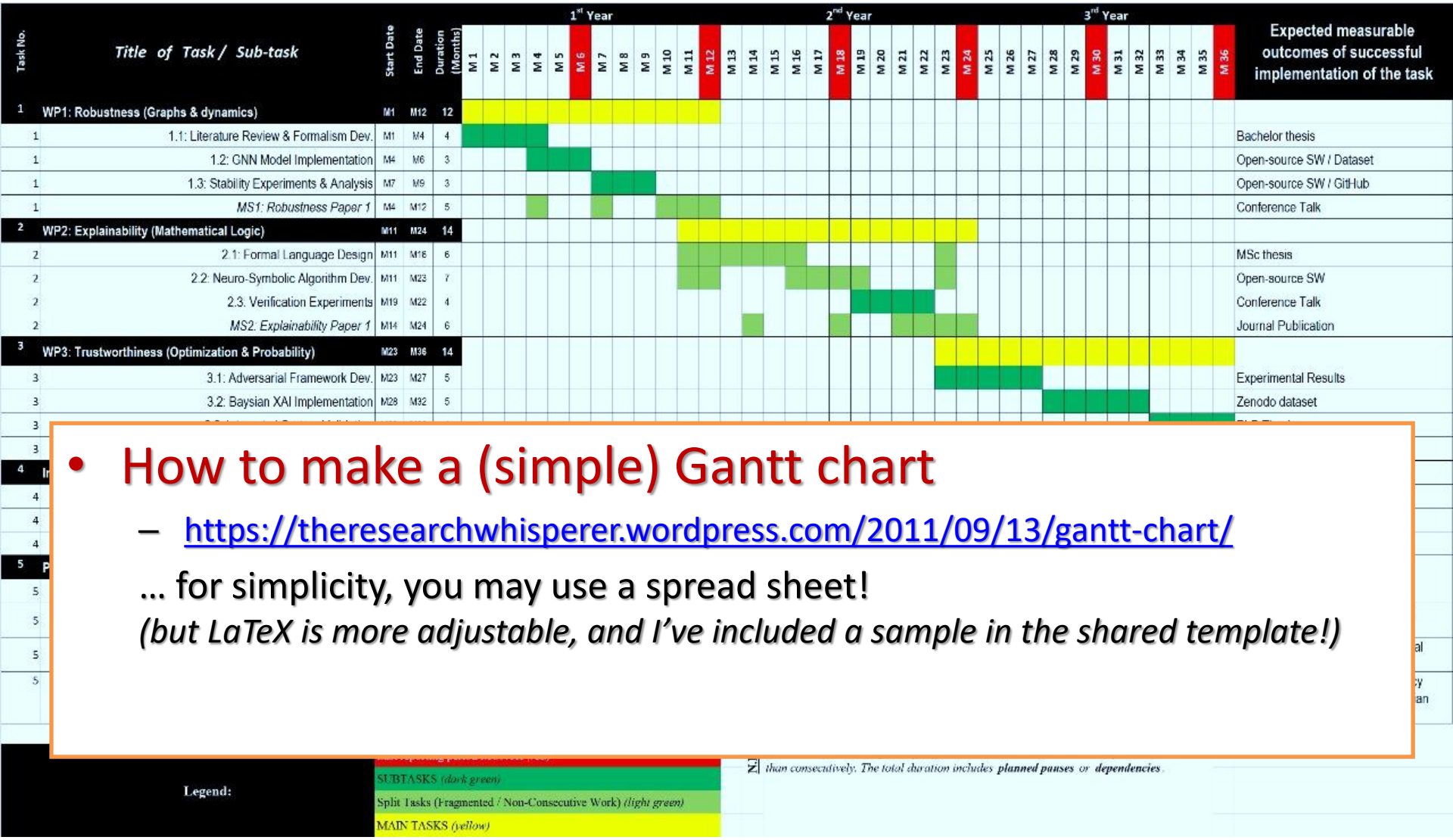
(general) Deliverables:

- Tools: We crossed this bridge…
- SMART
- Gantt Chart



Gantt Charts

https://en.wikipedia.org/wiki/Gantt_chart



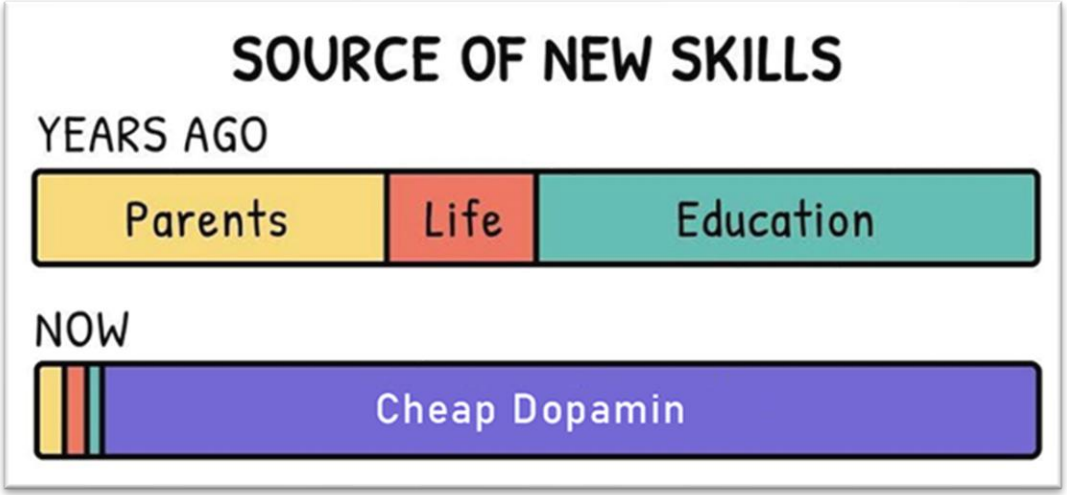
- How to make a (simple) Gantt chart
  - https://theresearchwhisperer.wordpress.com/2011/09/13/gantt-chart/

… for simplicity, you may use a spread sheet!

(but LaTeX is more adjustable, and I've included a sample in the shared template!)

Follow Up *(important)*

- We will use a shared folder (Google Drive).
- You'll create subfolders for *(described in the next slide)*
  - "**Composing**"
  - "**Sources**"
  - "**Dataset**"
  - "**Code**"
  - "Others"



Follow Up *(the subfolders)*

- "**Composing**":
  the most important folder with whatever you create <u>related to documentation</u>
  - such as docx, pptx, excel, LaTeX, etc.
- "**Sources**": folder with the references and tools you use.
  - ALWAYS rename the pdf-files by adding a serial number between two brackets
    - year of publication – then short key words of the title.
      - Example: `[03]-2016-SaliencyPaperAhmed`.pdf
- "**Dataset**": folder with links and descriptions of the working datasets.
  - Include samples in the folder, and
  - **Deliver** a copy of the dataset in our meeting!!!
- "**Code**": folder with all details and created items related to your implementation.
  - If you are you using an online editor (e.g., Jupiter), then always make updated copies for your work.
  - If you have a GitHub repository, you must share (and update) it.
- "Others": anything else, if at all!

Follow Up *(the meetings)*

- Weekly Meetings
  - Meeting Minutes

Date: Wednesday, 15/3/2017
Time: 4:30 PM
Location: C7.311 (Office)
Subject: 2nd Individual Meeting [5th Meeting (#7)]

	Point/Task	Action
1	Progress report	
2	Discuss problems with setting up development environment	Finish setting up the development environment
3	Discuss the backend implementation	Start implementing the backend components
4	Thesis	Start by writing the abstract, the introduction, and the academic survey so far

Progress Report

Week	Deliverables (Max Mark 10)	Quality (Max Mark 15)	Presentations (Max Mark 10)
1			
2			
3			
4			
5			
6			
7			

Follow Up *(the deliverables)*

- Weekly Meetings
  - Deliverables
    - Short Report
      - PDF or TeX
    - Short Presentation
      - 5 Min presentation (uploaded to "composing")
      - *(optional)* Add narrations/voice
    - Code results
      - Results that you like to share
      - N.B. Debug your code yourself!

Task 1

IMPORTANT !

- Compose a short description on your topic (1 page of text; LaTeX or PDF).
- Build a list of expected deliverables (5-8 points – weekly planned).
- Give a list of references used so far (at least 2).
- Give a list of possible further resources.

In total, max. 2 pages / Due the next meeting

1st Meeting !

Task 2

5 Min Presentation:

1. Overview of the re-phrased topic
2. Project Plan, including
   a. time plan,
   b. list of objectives/deliverables and
   c. Block diagram / visual abstract

Start NOW!

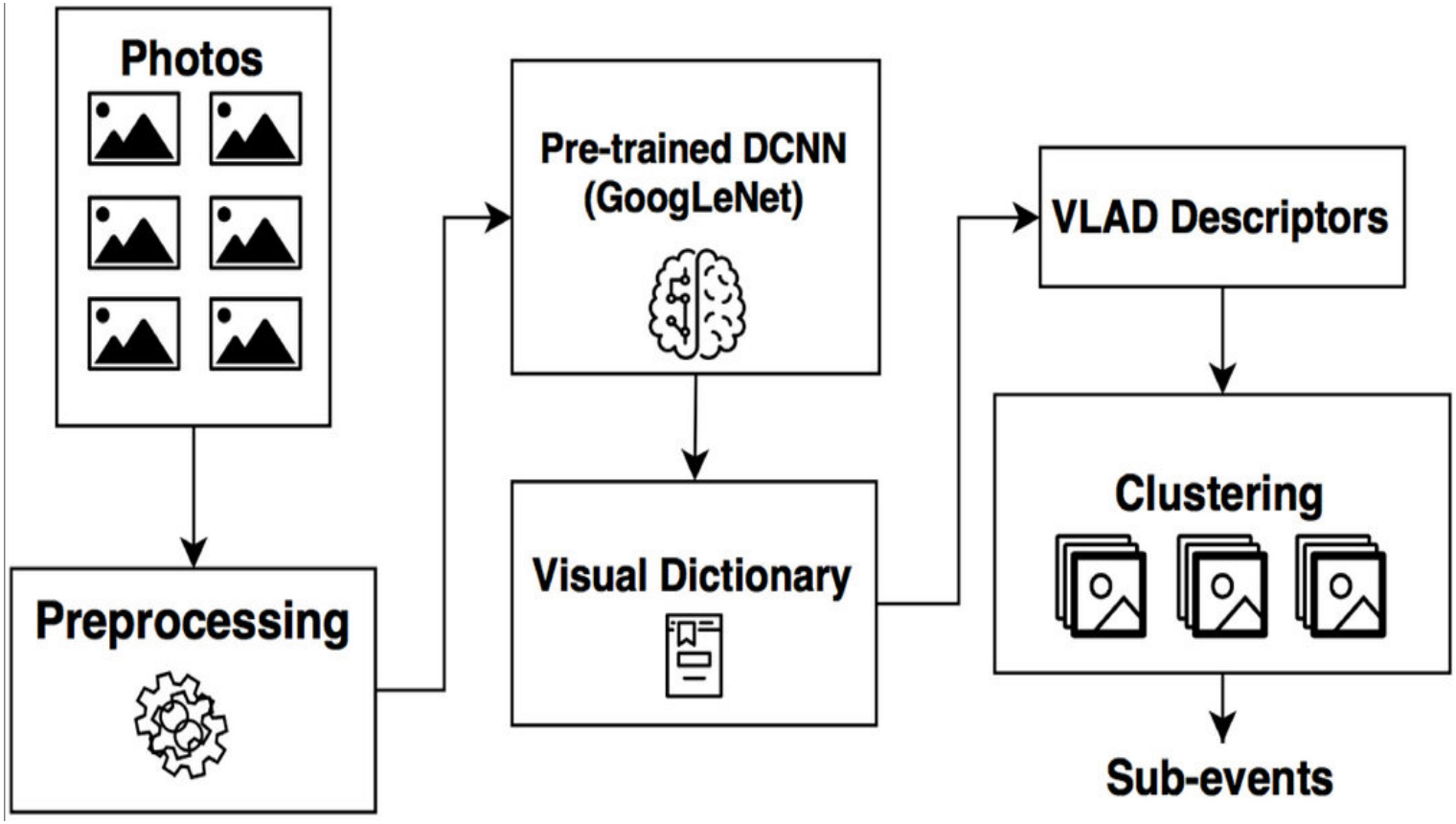
Interim Reports: (Report 1)

- Use LaTeX for both of your reports (extract from the thesis).
- First Report deadline: 19-March-2026
- First report includes:
  - Motivation and (finalized) well-definition of project problem,
  - Detailed information about your status
    - What is completed and what's missing (according to <u>your</u> plan)
  - **Literature review**: Summary of the papers related to your topic (use the ones you collected and read).
  - Your work & implementation plan

    ➔ use **block diagram**.

Follow Up *(the technicalities)*

- Block Diagram



Follow Up *(the technicalities)*

- A block-diagram is some sort of a flowchart.
- It summarizes your ideas and/or a process.
- They are important for the supervisor and the reviewer (and yourself).
- **Block-diagram** show graphically how the input data go through the deferent steps in the method to generate the output.
- You may create multiple block diagrams for the details of the project modules.

Interim Reports: (Report 2)

- Use latex for both of your reports (extract from the thesis).
- Second Report deadline: Thursday 07-May-2026
- Second report includes:
  - Completed **Chapters**
  - **Algorithm and implementation**
  - Results (even incomplete)
  - Updated block diagrams
- Keep in mind:
  - this report should closely resemble your final thesis!
  - further help is unlikely beyond revising/commenting on this report.

The FINAL Thesis

- Final Thesis deadline: 8-June-2026
- Content:
  - Title Page [Title, Student Name, Supervisor Name, Submission Date]
  - [Graphical] Abstract *(← this is where block diagrams will prove most useful!)*
  - Introduction Chapter
  - Literature Review Chapter
  - Report body [Methodology]
  - Results & Discussion
  - Conclusion
  - References
  - Don't forget page number (not necessarily the first page).
  - Use the provided GUC template
  - Apply language and plagiarism check!
  - Follow GUC Policy regarding Gen-AI tools.

Rehearsal (before defense)

- 15 min
- Introduce the topic.
- motivate the audience to why listen to your work.
- Go in details in the method.
- Show implementation details.
- Show advantages and disadvantages of the method you used to solve the problem.
- Show and analyze results.
- Leave the audience with a clear picture of your work, make them want to read your report.
- Test your skills, learn your drawbacks, get feedback, correct your path.

Deliverables: Presentation

- Presentations are a way for communication!
  - Talk, instead of read.
  - Stand up! Move around and move hands.
  - Make eye contact with your audience.
  - Don’t only look at one side of the room.
  - Don’t just read summary cards or whatever !!!!!
  - Watch your: Speed, Tone, and Volume!
  - Use laser pointer… Watch your hand move!
  - **Bring your (correct) adapter!**

Deliverables: Presentation

- Presentation Slides Style
  - Black text color on white background?
  - Each slide has a title?, slide number?
  - Group many slides under one title?
  - Introduce slides for outline to remind us about the previous and next topics?
  - A picture is worth…?

Deliverables: Presentation

- Presentation Slides Content
  - Title Page
  - Outline
  - Introduction: Motivation, Problem Statement, Current state of literature
  - Paper body: Challenges, Methodology, Results
  - Conclusion / Summary
  - References

  - Few text, more figures.
  - Each slide 1 - 2 min (long) on average.

Main Themes SS26: The Language of Trust

- The future of scientific research in AI is defined by Formalization and Representation. It involves turning vague concepts like "understanding" and "analogy" into rigorous structures that can be modeled and verified.
- The high-impact potential lies in creating systems that are hallucination-free by design—*systems that do not merely generate text that looks like math, but generate proofs that are math*.
- As the industry moves from chatbots to agents, the researchers who thrive will be the "Architects" who can build these provably understandable and reliable systems.

SS25 Clusters & Titles: NeSy Architects

(Sat./Mon./Tue.) Exact meeting times will be shared soon

ProjID:	Student Name	Topic
NeSy1	Nadine Abdelrahman Awny	The AI **Pinocchio** Detector: Automated Fact-Checking
NeSy2	Ali Hesham Abdelhady Ali Abdelrazek	**The Curriculum Architect**: Engineering the Perfect Learning Path
NeSy3	Rawan Osama	**The Honest Tutor**: Building "Glass-Box" AI in Education
NeSy4	*Khaled Mohamed Karam Alah*	*Multi-Sensory Search Engine: **The Cross-Modal Cartographer***
NeSy5	*Mohamed Adel Mahran Mohamed*	***The Visual Knowledge Brain***
NeSy6	*Mohamed Karim Nabil*	***The Algorithmic Orchestrator**: Policy-Driven Graph Optimization*
NeSy7	*Omar Hazem Shaaban*	*The Modern **Rosetta Stone** of Math: DeepMind-Style Truth Scanner*

SS25 Clusters & Titles: Cognitive Auditors

(Sat./Mon./Tue.) Exact meeting times will be shared soon

ProjID:	Student Name	Topic
Cog1	Habiba Mohamed Samy Elhabiby	The "**Terms of Service**" Legal Checker
Cog2	Mohamed Ihab Seifeldin Zakaria	**The Prerequisite Pathfinder**
Cog3	Rawan Hossam Kamal	Modeling of Social Dynamics: Sternberg's "**Love Quantifier**"
Cog4	Abdelrahman Moh. Abdelrahman	**System-2 Reasoning**: The Wise Agent
Cog5	Omar Moh. Moh. Kamal Eldafrawy	System-2 Planning: The **Lookahead** Agent
Cog6	Alya Ahmed Hamdy Ibrahim	The Empathy Auditor: The **ELIZA** Effect 2026
Cog7	Ahmed Tamer Moh. Waheed Kasem	Multi-Agent Pathfinding: The **Traffic Controller**
Cog8	Adham Ashraf Ismail Abdelaal	System-2 Planning: The **Deadlock Detective**
Cog9	Adel Ashraf Mohamed Kamel Emam	**The Logic Advocate Tutor**

SS25 Clusters & Titles: Human-Centric

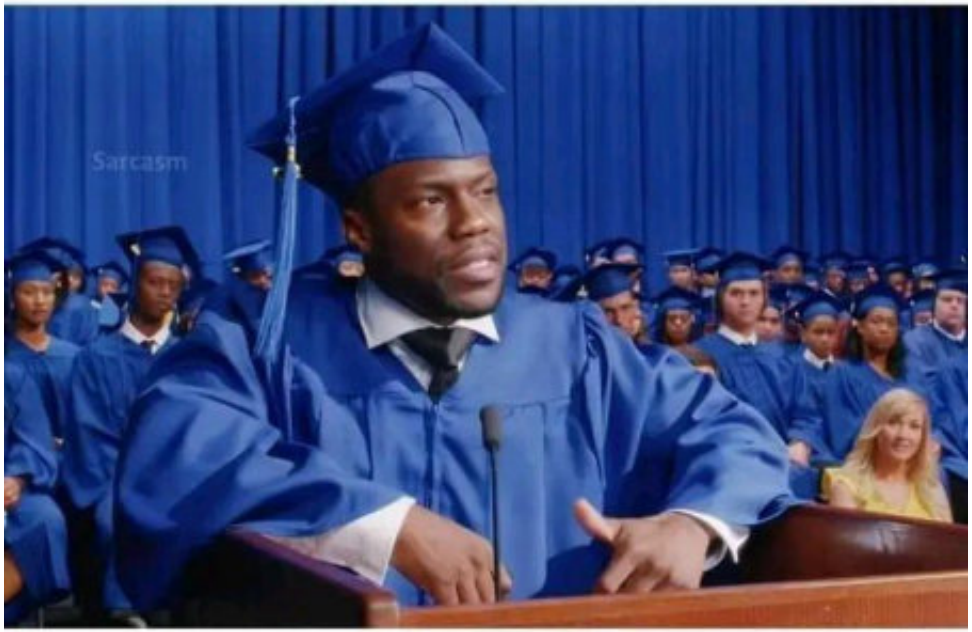
(Sat./Mon./Tue.) Exact meeting times will be shared soon

ProjID:	Student Name	Topic
GenH1	Mahmoud Ashraf Anwar	Sketch-based **Heritage Restoration**
GenH2	Mennatallah Moh. Abdelfatah	**The Kinematic Calligrapher** *(In Collaboration with Dr Yomna Hassan)*
GenH3	Ahmed Walid Monir Ahmed	**The Algorithmic Complexity Visualizer** *(In Collaboration with Dr Yomna Hassan)*
GenH4	Ahmed Hany Ibrahim Elhussieny	The **Dysgraphia** Detective *(In Collaboration with Dr Yomna Hassan)*
GenH5	Mostafa Rabeea Marzouk	The Digital **Handwriting Scorer** *(In Collaboration with Dr Yomna Hassan)*

DEFENSE

Graduation speech:

I would like to thank ...



The Internet, Google, Wikipedia,

Microsoft office, and Copy/Paste.

The Student’s Commitment:

I'm beginning my learning in Bachelor Thesis, and I know learning can be hard, but I have the patience, determination, and discipline to reach my goals.

When I'm stuck, I'll find a solution. I’ll do my best to:

- Take a breath, ☺
- Break it down,
- Search seriously for a serious help,
- Research related literature (not using general-purpose LLMs),
- Stay positive (there is always Plan B),
- Talk through it.

I commit to completing this course:

Yes / No

SOURCES TO START WITH

There is no way to give an exhaustive list of resources!
The list (of today) will constantly be updated:

- **2026-02-12: Orientation Day SS26**
  - Open Access to a large collection of resources
  - My Shared Drive for SS26

Contact:

Dr. **Ahmed M. H. Abdelfattah**

Office: C7.207

Clarity is the kindest rudeness.

Faculty of Media Engineering and Technology,
German University in Cairo (GUC)

Email: ahmed.abdelfattah@guc.edu.eg
Link: https://www.linkedin.com/in/ahmed-abdelfattah-371b273a/

HOW TO READ A PAPER

How to read a paper

- Paper types:
  - Comparison between different methods,
  - Novel methods (and a comparison),
  - Combination of methods,
  - Testing existing methods with new data / application.
  - Position paper?
  - Etc.

How to read a paper

- Recognize:
  - Motivation, Problem, Application, Methods,
  - Data, Implementation tools
  - The paper tells how the author used these methods to propose a solution to this problem within the domain of this Application.
- Recognize:
  - Novelty, Contribution, Correctness, Limitations, Clarity.
- Find out:
  - Keywords, Terminology and acronyms

How to read a paper

- Build: **Block-diagram** that show graphically how the input data go through the different steps in the method to generate the output.

- Block-diagram is some sort of a flowchart.

How to read a paper

- Three-pass method for reading research papers
  - The first pass gives you a general idea about the paper. The second pass lets you grasp the paper's content, but not its details. The third pass helps you understand the paper in depth.
  - The 1st pass:
    - Carefully read the title, abstract, and introduction
    - Read the section and sub-section headings, but ignore everything else
    - Read the conclusions
    - Glance over the references, mentally ticking off the ones you've already read

How to Read a Paper S. Keshav David R. Cheriton School of Computer Science, University of Waterloo Waterloo, ON, Canada, keshav@uwaterloo.ca, http://ccr.sigcomm.org/online/files/p83-keshavA.pdf

How to read a paper

- Three-pass method for reading research papers
  - The 2nd pass: read the paper with greater care, but ignore details such as proofs
    - Look carefully at the figures, diagrams, and graphs. Are the axes properly labeled? Are results shown with error bars, so that conclusions are statistically significant?
    - Remember to mark relevant unread references for further reading (this is a good way to learn more about the background of the paper).
  - The 3rd Pass: attempt to virtually re-implement the paper
    - Great attention to detail
    - you should think about how you yourself would present a particular idea.

How to Read a Paper S. Keshav David R. Cheriton School of Computer Science, University of Waterloo Waterloo, ON, Canada, keshav@uwaterloo.ca, http://ccr.sigcomm.org/online/files/p83-keshavA.pdf